

EX.NO:7 IMPLEMENTATION OF A MULTI-THREAD PROGRAM

Program:

```
import java.util.Arrays;
import java.util.Random;

class ArrayGenerator implements Runnable {
    private int[] arr;
    public ArrayGenerator(int[] arr) {
        this.arr = arr;
    }

    @Override
    public void run() {
        Random rand = new Random();
        for (int i = 0; i < arr.length; i++) {
            arr[i] = rand.nextInt(100); // random numbers between 0-99
        }
        System.out.println("Generated Array: " + Arrays.toString(arr));
    }
}

class AscendingSorter implements Runnable {
    private int[] arr;
    public AscendingSorter(int[] arr) {
        this.arr = arr;
    }

    @Override
    public void run() {
        int[] copy = Arrays.copyOf(arr, arr.length);
        Arrays.sort(copy);
        System.out.println("Ascending Order: " + Arrays.toString(copy));
    }
}

class DescendingSorter implements Runnable {
    private int[] arr;
    public DescendingSorter(int[] arr) {
        this.arr = arr;
    }

    @Override
    public void run() {
        int[] copy = Arrays.copyOf(arr, arr.length);
        Arrays.sort(copy);
        // Reverse for descending
        for (int i = 0; i < copy.length / 2; i++) {
            int temp = copy[i];
            copy[i] = copy[copy.length - 1 - i];
            copy[copy.length - 1 - i] = temp;
        }
        System.out.println("Descending Order: " + Arrays.toString(copy));
    }
}

public class MultiThreadArray {
    public static void main(String[] args) throws InterruptedException {
        int[] arr = new int[10]; // array of size 10
        Thread generator = new Thread(new ArrayGenerator(arr));
        generator.start();
        generator.join(); // wait until array is generated
        Thread ascSorter = new Thread(new AscendingSorter(arr));
        Thread descSorter = new Thread(new DescendingSorter(arr));
        ascSorter.start();
        descSorter.start();
        ascSorter.join();
    }
}
```

```
descSorter.join();  
}  
}
```

Output:

```
Generated Array: [45, 12, 78, 34, 23, 89, 56, 10, 67, 90]  
Ascending Order: [10, 12, 23, 34, 45, 56, 67, 78, 89, 90]  
Descending Order: [90, 89, 78, 67, 56, 45, 34, 23, 12, 10]
```